Random Variables and Probability Distributions

My Statistics and Probability reviewer. Random variables sound scary, but they just attach numbers to the results of a chance experiment so we can compute with them.

What is a random variable?

A random variable is a function that assigns a real number to each outcome in the sample space of an experiment. We usually name it with a capital letter like X, and the specific values it takes with lowercase letters like x.

Example: toss two coins. The sample space is {HH, HT, TH, TT}. If X is the number of heads, then HH gives 2, HT and TH each give 1, and TT gives 0. So X can take the values 0, 1 and 2.

Discrete vs continuous

- A discrete random variable has a countable set of possible values, usually from counting. Examples: the number of jeepneys passing a corner in ten minutes, the number of correct answers on a quiz, the number of siblings a student has.
- A continuous random variable can take any value in an interval, usually from measuring. Examples: the height of a Grade 11 student, the time it takes to commute to school, the amount of rainfall during a typhoon.

Quick test: if you can list the values (even forever, like 0, 1, 2, 3, ...), it is discrete.

Probability distribution and the PMF

A probability distribution tells us every value a random variable can take and how likely each one is. For a discrete random variable, this is described by the probability mass function, or PMF, written P(X = x) or simply P(x). It can be shown as a table, a graph (histogram), or a formula.

A PMF is valid only if both rules hold:

- Each probability is between 0 and 1 inclusive: 0 <= P(x) <= 1.
- All the probabilities add up to exactly 1: sum of P(x) = 1.

If either rule fails, it is not a probability distribution. For example, a table with probabilities 0.5, 0.4 and 0.2 is invalid because the sum is 1.1.

For continuous random variables we use a probability density function instead, and probabilities are areas under the curve. For a continuous variable, the probability of any single exact value is 0.

Mean or expected value

The mean of a discrete random variable, also called its expected value, is the long-run average you would get if you repeated the experiment many times. It is a weighted average, where each value is weighted by its probability.

- Formula: mu = E(X) = sum of [x times P(x)]

The expected value does not need to be one of the possible values of X. For example, the expected number of heads in two coin tosses is 1, but in other problems you might get something like 1.7.

Variance and standard deviation

Variance measures how spread out the values are around the mean.

- Definition formula: sigma^2 = sum of [(x - mu)^2 times P(x)]
- Shortcut formula: sigma^2 = sum of [x^2 times P(x)] - mu^2

The standard deviation is the positive square root of the variance: sigma = sqrt(sigma^2). It is in the same units as X, which makes it easier to interpret than the variance. A bigger standard deviation means the outcomes are more scattered; a smaller one means they cluster close to the mean.

Fully worked example

A sari-sari store owner in Batangas tracks X, the number of cases of softdrinks delivered on a given day. Her records give this distribution:

- x = 0, P(x) = 0.10
- x = 1, P(x) = 0.30
- x = 2, P(x) = 0.40
- x = 3, P(x) = 0.20

Step 1. Check that it is valid. Each probability is between 0 and 1, and 0.10 + 0.30 + 0.40 + 0.20 = 1.00. Valid.

Step 2. Find the mean.

- 0(0.10) = 0
- 1(0.30) = 0.30
- 2(0.40) = 0.80
- 3(0.20) = 0.60
- mu = 0 + 0.30 + 0.80 + 0.60 = 1.70

So on average, 1.7 cases are delivered per day.

Step 3. Find the sum of x^2 times P(x).

- 0^2(0.10) = 0
- 1^2(0.30) = 0.30
- 2^2(0.40) = 1.60
- 3^2(0.20) = 1.80
- Total = 3.70

Step 4. Use the shortcut formula for variance.

- sigma^2 = 3.70 - (1.70)^2 = 3.70 - 2.89 = 0.81

Step 5. Take the square root.

- sigma = sqrt(0.81) = 0.9

Interpretation: deliveries usually fall within about 0.9 cases of the average of 1.7 cases.## Quick check

- Q: Is the number of text messages you receive in a day discrete or continuous? A: Discrete, because it comes from counting.
- Q: What two conditions must a probability mass function satisfy? A: Each P(x) is from 0 to 1, and all P(x) values add up to 1.
- Q: What is another name for the mean of a random variable? A: The expected value, E(X).
- Q: If the variance of X is 0.81, what is its standard deviation? A: 0.9.

- Q: Give the shortcut formula for the variance of a discrete random variable. A: sigma^2 = sum of [x^2 times P(x)] - mu^2.